Dr. Ram Manohar Lohia Avadh University, Ayodhya

MINOR PROJECT ON

"EXPLORATORY DATA ANALYSIS"

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CERTIFICATE

Elite Softwares

TO WHOMSOEVER IT MAY CONCERN

INTERNSHIP CERTIFICATE

This is to certify that **Shivam Shukla** has successfully completed their internship as a **"Software Engineer Intern"** in a technology **Web Development, Python and DJango Framework"** under the assistance and guidance in connection with his/her Industrial Internship at Elite Softwares, Pune.

Duration of this Internship was from **1st / November / 2025 to 1st / January / 2026**

He/she was found sincere & hard working during this tenure.

We wish him/her all the best for his/her future endeavors.

Issue Date: 9th / January / 2026

Elite Softwares Pune

Best Regards,

Swami Panjala
Founder and CEO
Elite Softwares, Pune

MSME – UDYAM No. – MH-26-0565456

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I feel extremely privileged to express my deep sense of gratitude and regard to my supervisor
MR. PAWAN KUMAR, for their expert supervision, valuable and critical suggestions. I will always remain indebted to them for their undivided attention pointed thrust and constant encouragement in my current endeavour.

At last I would like to thank all those who could not find a separate mention but have helped me directly or indirectly.

SHIVAM SHUKLA
23060100010070

TABLE OF CONTENT

S. No.	Particulars	Page No.
1	Front Page	i
2	Certificate	ii
3	Acknowledgement	iii
4	Abstract	iv
5	Introduction	1
6	Domain Description	3
7	Problem Statement	5
8	Objectives of the Study	7
9	Scope of the Work	8
10	Literature Review	10
11	Tools and Technologies	12
12	Data Understanding	14
13	Data Preprocessing	17
14	Exploratory Data Analysis	20
15	Data Visualization	23
16	Statistical Analysis	25
17	Business Insights	27
18	Code Detail	29
19	Recommendations	32
20	System Environment	34
21	Limitations	36
22	Future Scope	37
23	Conclusion	39
24	References	41

Abstract

The Diwali Sales Analysis project focuses on examining festive season retail transaction data to identify patterns in customer purchasing behavior, demographic influence, and product performance using data analytics techniques. Diwali, being one of the most economically significant festivals in India, generates a substantial surge in retail sales across multiple product categories. Understanding consumer behavior during this period is essential for businesses to optimize marketing strategies, inventory planning, and revenue generation.

This study utilizes a dataset containing over 11,000 transaction records comprising demographic attributes such as gender, age, marital status, occupation, and state, along with transactional attributes including product category, number of orders, and purchase amount. The project applies systematic data preprocessing techniques to clean and structure the dataset by removing null values, correcting data types, and eliminating redundant information.

Exploratory Data Analysis (EDA) is performed using Python programming language and libraries such as Pandas, NumPy, Matplotlib, and Seaborn. Statistical methods including aggregation, grouping, and descriptive analysis are employed to identify trends and patterns in festive purchasing behavior. Visualization techniques are used to represent demographic distributions, regional sales contribution, and product category performance in an interpretable graphical format.

The analysis reveals that female customers contribute a higher overall revenue share during the Diwali season, and individuals within the 26–35 age group demonstrate the highest purchasing activity. State-wise evaluation indicates regional concentration of festive sales, with certain economically strong states contributing significantly higher revenue. Occupational analysis further highlights the relationship between professional stability and purchasing power.

The findings emphasize the importance of data-driven decision-making in retail management. By transforming raw transactional data into actionable insights, the study demonstrates how businesses can enhance festive sales strategies, improve customer targeting, and optimize operational efficiency. Overall, this project illustrates the practical application of data analytics in understanding seasonal retail behavior and supporting strategic business growth.

Introduction

In the contemporary era of digital transformation, data has emerged as one of the most valuable strategic assets for organizations across industries. Every transaction, customer interaction, online purchase, and store visit generates data that, when analyzed appropriately, can reveal significant insights about consumer behavior and business performance.

The retail sector, in particular, relies heavily on data analytics to understand trends, forecast demand, and improve customer engagement strategies. In this highly competitive environment, organizations that effectively utilize data analytics gain a substantial advantage over those that rely solely on intuition or traditional business practices.

Festivals play a crucial role in driving retail sales, especially in India. Among all festivals, Diwali holds exceptional economic significance. Diwali is not merely a religious celebration but also a period characterized by large-scale consumer spending across various product categories such as electronics, clothing, jewelry, home décor, gifts, kitchen appliances, and luxury goods. Retailers and e-commerce platforms experience a surge in customer demand during this festive period, often accounting for a significant percentage of their annual revenue. Due to this seasonal spike, businesses need accurate analysis of festive sales patterns to make strategic decisions regarding inventory

management, marketing campaigns, and promotional offers.
Despite the importance of festive sales, many businesses fail to conduct systematic analytical studies on their transaction data. They may record the sales but do not deeply analyze customer demographics, spending patterns, regional contributions, or product performance metrics. This creates a gap between data collection and data-driven decision-making. The Diwali Sales Analysis project is developed to bridge this gap by applying structured data analysis techniques using Python programming tools.

The project focuses on analyzing a dataset containing over eleven thousand records of customer transactions during the Diwali season. It includes multiple attributes such as gender, age group, marital status, occupation, state, product category, number of orders, and purchase amount. By performing data cleaning, preprocessing, exploratory data analysis, and visualization, the study aims to uncover meaningful patterns that can provide actionable insights for business enhancement. The objective of this analysis is not only academic fulfillment but also practical demonstration of how analytical tools such as Pandas, NumPy, Matplotlib, and Seaborn can be used in real-world retail environments. This study demonstrates the transformation of raw transactional data into strategic business intelligence.

DOMAIN DESCRIPTION

The Diwali Sales Analysis project falls under the broader academic and professional domains of Data Analytics, Retail Analytics, and Business Intelligence. Data analytics is a multidisciplinary field that combines statistics, mathematics, programming, and domain knowledge to extract actionable information from raw datasets. It involves techniques such as data cleaning, transformation, visualization, and interpretation to support decision-making.

Retail analytics is a specialized branch of data analytics that focuses specifically on consumer purchasing behavior and sales trends within the retail industry. In today's highly dynamic marketplace, retailers collect vast volumes of transactional data across offline stores and online platforms. Retail analytics helps organizations examine patterns such as seasonal demand fluctuations, product popularity, regional purchasing differences, and customer segmentation.

Business intelligence is closely related to retail analytics and refers to the use of software tools and methodologies that transform raw data into meaningful and useful information for strategic planning. Through dashboards, data visualization, and reporting tools, business intelligence systems provide insights into profitability, revenue growth, and operational efficiency.

The festive sales domain is particularly important in emerging economies such as India, where cultural festivals significantly influence consumer expenditure. Consumer psychology during festive seasons differs from regular purchasing behavior, as emotional, social, and cultural factors encourage higher spending.

This project integrates analytical concepts from all these domains to evaluate festive sales performance systematically. By applying computational tools to retail data, the project contributes to data-driven business practices and reinforces the importance of analytics in modern retail management.

PROBLEM STATEMENT

Retail companies experience substantial revenue growth during the Diwali season, yet many fail to leverage transactional data to understand deeper customer and sales patterns. The absence of structured data analysis results in inefficient planning and missed optimization opportunities. Businesses often make assumptions about customer preferences without solid evidence derived from data.

The major problem addressed in this project is the lack of systematic evaluation of festive sales data. Retailers are often unable to determine which demographic group contributes the most revenue, which state performs best, which occupation group has the highest purchasing capacity, or which product categories generate maximum profits. This leads to ineffective marketing strategies, uneven inventory allocation, and missed profit potential.

Furthermore, without demographic segmentation, businesses cannot personalize marketing efforts or design targeted promotional campaigns. Inventory may be overstocked in low-demand regions while high-demand areas suffer shortages. This inefficiency increases operational costs and reduces overall profitability.

The Diwali Sales Analysis project aims to address these issues by conducting comprehensive exploratory data analysis. It attempts to identify patterns in customer demographics, product categories, and regional performance to provide meaningful insights that can assist retail businesses in optimizing their strategies.

OBJECTIVES OF THE STUDY

The primary objective of this study is to analyze Diwali festive sales data to extract valuable insights using data analytics tools. However, this broad objective is supported by several specific sub-objectives designed to ensure systematic analysis and interpretation.

One important objective is to perform data cleaning and preprocessing to remove inconsistencies, missing values, and redundant columns. Clean data ensures reliable results and accurate calculations.

Another objective is to examine gender-based purchase behavior and understand how male and female consumers differ in spending patterns during the festive season. The study also aims to analyze age group distributions to identify the most economically active demographic segment.

State-wise sales contribution analysis is another key objective. By aggregating revenue by region, the project identifies geographic concentration of festive spending.

Occupation-based analysis helps determine whether certain professional groups contribute more to festive sales due to higher income levels or purchasing preferences.

Finally, the project aims to interpret results through graphical visualization, enhancing clarity for presentation and reporting purposes

SCOPE OF THE WORK

The scope of the Diwali Sales Analysis project is defined around the systematic exploration and interpretation of festive season retail data using structured data analytics techniques. This project primarily focuses on understanding customer purchase behavior, demographic influence, product category performance, and regional sales contribution during the Diwali festival. The dataset used in the study contains over eleven thousand transaction records, representing customer purchases made during the festive period.

The scope includes comprehensive data cleaning, preprocessing, exploratory data analysis, and visualization of sales patterns. Through preprocessing techniques, missing values are identified and removed, redundant columns are eliminated, and data types are standardized to ensure consistency. After cleaning, descriptive statistics and aggregation operations are applied to determine metrics such as average purchase amount, total sales revenue by demographic category, and order frequency.

The project emphasizes exploratory data analysis rather than predictive modeling. It focuses on understanding existing patterns in historical data rather than forecasting future sales performance. Visualization techniques such as bar charts and count plots are used to enhance interpretability.

Although the analysis provides strong insight into festive purchasing behavior, it is restricted to a single dataset representing one festive season. It does not incorporate multi-year comparison, predictive algorithms, or real-time business dashboards. Nevertheless, within its defined scope, the project successfully demonstrates the application of Python-based data analytics in retail analysis.

LITERATURE REVIEW

Over the past decade, retail analytics has gained significant importance in both academic research and commercial practice. Numerous studies have examined the impact of seasonal and festive events on consumer purchasing patterns. Research in marketing analytics suggests that cultural and emotional factors strongly influence consumer spending during festivals, often leading to higher transaction volumes compared to regular shopping periods.

Studies in demographic segmentation highlight that gender, age, income level, and occupation significantly affect purchasing decisions. For instance, research findings commonly show that economically stable age groups between 25 to 40 years demonstrate higher discretionary spending. Similarly, several retail behavior studies indicate that women often dominate purchasing in categories such as clothing, home décor, and gifts, particularly during festive occasions.

Geographical sales distribution analysis is another widely explored area in retail research. Scholars emphasize that urbanized regions with higher economic activity contribute disproportionately higher revenue during festive seasons. Researchers also stress the importance of visual analytics and statistical summaries in understanding transactional datasets.

Python has emerged as a prominent tool in data analytics research due to its flexibility and robust libraries. Academic literature frequently documents the use of Pandas for data manipulation, NumPy for numerical operations, and Matplotlib and Seaborn for statistical visualization.

The Diwali Sales Analysis project aligns with these research findings by applying exploratory data analysis to identify demographic and regional trends in festive season retail data. It reinforces existing theories on customer segmentation and highlights the value of data-driven decision-making.

TOOLS AND TECHNOLOGIES

The execution of this project relies on a combination of modern programming tools and analytical libraries designed specifically for data manipulation and visualization. Python serves as the core programming language due to its simplicity, readability, and extensive ecosystem of data science libraries. Its open-source nature makes it accessible for academic and professional use.

Pandas plays a central role in handling tabular data. It provides powerful DataFrame structures that allow efficient filtering, grouping, aggregation, and statistical operations. Through functions such as groupby(), describe(), and value_counts(), large transactional datasets can be summarized effectively.

NumPy supports numerical computations and enables mathematical operations on arrays, improving computational efficiency when performing aggregate calculations.

Matplotlib is utilized for creating graphical representations such as bar charts, while Seaborn enhances visualization aesthetics and provides built-in support for statistical plotting. These visualization tools transform numerical summaries into intuitive graphical insights.

Jupyter Notebook is used as the development environment, offering an interactive interface that combines code execution, visualization, and documentation in a single platform. This environment enhances experimentation and iterative analysis.

Together, these tools create a comprehensive analytical environment capable of handling large retail datasets and generating meaningful insights.

DATA UNDERSTANDING

Data Understanding is one of the most critical stages in the data analytics lifecycle. Before performing any computation, visualization, or modeling, it is essential to thoroughly understand the dataset's structure, attributes, relationships, and limitations. In the Diwali Sales Analysis project, the dataset consists of 11,251 rows and 15 columns, where each row represents an individual customer transaction during the Diwali festive season.

The dataset includes demographic attributes such as Gender, Age, Age Group, Marital Status, State, and Occupation. It also includes transactional attributes such as Product Category, Orders, and Amount. These attributes collectively allow multidimensional analysis of customer purchase behavior.

The first step in data understanding involved inspecting the dataset using functions such as .info(), .head(), and .describe() in Pandas. The .info() function helped determine the number of non-null values and the data type of each column. It revealed that certain columns such as "Status" and "unnamed1" contained mostly null values and did not contribute any meaningful information.

The .describe() function provided statistical summaries including count, mean, standard deviation, minimum, maximum, and quartile values for

numeric columns. For example, analyzing the Amount column revealed the distribution of customer spending. Observing maximum and minimum purchase values allowed identification of spending range. Additionally, categorical distribution was studied using value counts. This helped determine which gender category had more customers and which states had higher representation in the dataset.

Understanding data structure ensured that analysis would be logically correct and statistically valid. Without this foundational step, further aggregation or visualization might lead to incorrect interpretations.

DATA PREPROCESSING

Data preprocessing is a crucial process that transforms raw, unstructured data into a clean and usable format suitable for analysis. In real-world datasets, inconsistencies such as missing values, redundant columns, incorrect data types, and duplicate entries are common. The Diwali Sales dataset required systematic preprocessing to ensure analytical accuracy.

The initial dataset inspection identified columns such as “Status” and “unnamed1,” which contained entirely null values. Retaining such columns could increase computational overhead and create confusion. Therefore, these columns were removed using the .drop() function in Pandas.

Missing values in the Amount column were detected using .isnull().sum(). Since financial calculations rely heavily on the accuracy of purchase amounts, it was essential to remove records where the transaction amount was missing. The .dropna() function was used to eliminate incomplete rows.

Another important preprocessing step involved correcting data types. The Amount column was converted into integer format to ensure consistency in aggregation operations such as summation and averaging. If stored as object type, mathematical operations could produce errors.

Duplicate entries were checked to ensure no transaction was counted twice. Data normalization and consistency checks were also performed to ensure uniform formatting in categorical variables such as gender and occupation.

Through preprocessing, the dataset transitioned from raw transactional data into a clean, structured format ready for advanced analytical operations.

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is the process of discovering patterns, relationships, and trends in data using statistical and visualization methods. Unlike predictive modeling, EDA focuses on understanding the underlying structure of historical data.

In this project, EDA began with demographic-based grouping. Using Pandas groupby() function, total purchase amount was aggregated across gender categories. The results showed that female customers contributed a significantly higher overall revenue compared to male customers.

Age group analysis was performed to evaluate which age demographic spent the most during Diwali. Aggregation revealed that individuals in the 26–35 age group contributed the highest revenue. This suggests that financially stable young professionals dominate festive purchasing behavior.

Occupation-based grouping identified high-spending professional sectors such as IT, Healthcare, and Government services. This insight links economic stability to purchasing behavior.

State-wise analysis was conducted by aggregating revenue by state. Maharashtra emerged as the highest revenue contributor, indicating strong regional purchasing activity.

EDA provided structured insights that transform raw transaction rows into strategic interpretations.

DATA VISUALIZATION

Data visualization serves as a bridge between complex numerical analysis and human interpretation. In this project, visualization techniques were applied using Matplotlib and Seaborn libraries to represent aggregated findings graphically.

Bar charts were generated to compare gender-based revenue contributions. These graphs visually confirmed statistical results, making differences immediately observable.

Count plots were used to examine customer frequency distribution across age groups. This allowed understanding whether high revenue came from large customer volume or high individual spending.

State-wise bar charts visually displayed regional disparity in festive sales. Visualization enhanced interpretability for presentation and reporting.

Proper axis labeling, title specification, and rotation of categorical names improved readability.

Visualization does not merely beautify data; it strengthens analytical interpretation by converting raw numbers into meaningful patterns.

STATISTICAL ANALYSIS

Statistical analysis provides quantitative validation to insights derived from EDA. Measures such as mean, median, variance, and standard deviation were computed.

The mean purchase amount was approximately ₹9453, indicating moderate festive spending per customer. The maximum recorded transaction exceeded ₹23,000, indicating presence of high-value buyers.

Standard deviation helped measure variability in purchase amount. High variance suggests diverse purchasing power among customers. Quartile analysis identified distribution spread and spending concentration.

These statistical metrics provide mathematical support to graphical findings.

BUSINESS INSIGHTS

The insights derived from analysis have direct business implications. Female customers represent the highest revenue-generating segment, suggesting targeted festive campaigns toward women can increase profitability.

The 26–35 age demographic indicates strong purchasing power, likely due to financial independence.

Geographical concentration in Maharashtra suggests logistics and marketing budgets should prioritize high-performing states. Occupation-based segmentation allows customized offers tailored to professional sectors.

These insights enhance strategic planning and competitive advantage.

CODE DETAIL

Diwali Sales Analysis

February 25, 2026

EXPLORATORY DATA ANALYSIS

TITLE: Diwali Sales Analysis

PROBLEM STATEMENT

During the Diwali festival season, a large amount of sales data is generated across different customers, products, and regions.

Businesses often struggle to understand customer purchasing behavior and identify key factors influencing sales performance.

Lack of proper data analysis makes it difficult to recognize high-revenue product categories and profitable customer segments.

Inefficient insights can lead to poor marketing decisions and ineffective inventory planning.

This analysis aims to systematically study Diwali sales data using data analytics techniques to uncover trends, patterns, and actionable insights for improving festive sales strategies.

SUBMITTED BY

NAME : SHIVAM SHUKLA ROLL NO : 23060100010070 CLASS : BCA 5TH SEM COLLEGE : DR. RAM MANOHAR LOHIYA AWADH UNIVERSITY AYODHYA

1.Data Understanding

importing data and libraries

[3]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[3]:
```
df = pd.read_csv('Diwali Sales Data.csv', encoding= 'unicode_escape')
```

[4]:
```
df.shape
```

[4]:
```
(11251, 15)
```

[5]:
```
df.head
```

[5]:
```
<bound method NDFrame.head of        User_ID    Cust_name Product_ID Gender Age
Group  Age  Marital_Status  \
```

```
0      1002903    Sanskriti  P00125942      F     26-35   28               0
1      1000732       Kartik  P00110942      F     26-35   35               1
2      1001990        Bindu  P00118542      F     26-35   35               1
3      1001425       Sudevi  P00237842      M      0-17   16               0
4      1000588         Joni  P00057942      M     26-35   28               1
…        …          …        …    …       …  …             …
11246  1000695      Manning  P00296942      M     18-25   19               1
11247  1004089  Reichenbach  P00171342      M     26-35   33               0
11248  1001209        Oshin  P00201342      F     36-45   40               0
11249  1004023       Noonan  P00059442      M     36-45   37               0
11250  1002744      Brumley  P00281742      F     18-25   19               0

                State      Zone       Occupation Product_Category  Orders  \
0         Maharashtra   Western       Healthcare             Auto       1
1      Andhra Pradesh  Southern             Govt             Auto       3
2       Uttar Pradesh   Central       Automobile             Auto       3
3           Karnataka  Southern     Construction             Auto       2
4             Gujarat   Western  Food Processing             Auto       2
…               …       …               …               …     …
11246     Maharashtra   Western         Chemical           Office       4
11247         Haryana  Northern       Healthcare       Veterinary       3
11248  Madhya Pradesh   Central          Textile           Office       4
11249       Karnataka  Southern      Agriculture           Office       3
11250     Maharashtra   Western       Healthcare           Office       3

         Amount  Status  unnamed1
0      23952.0     NaN       NaN
1      23934.0     NaN       NaN
2      23924.0     NaN       NaN
3      23912.0     NaN       NaN
4      23877.0     NaN       NaN
…        …     …        …
11246    370.0     NaN       NaN
11247    367.0     NaN       NaN
11248    213.0     NaN       NaN
11249    206.0     NaN       NaN
11250    188.0     NaN       NaN

[11251 rows x 15 columns]>
```

[6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   User_ID           11251 non-null  int64
```

```
 1   Cust_name         11251 non-null  object
 2   Product_ID        11251 non-null  object
 3   Gender            11251 non-null  object
 4   Age Group         11251 non-null  object
 5   Age               11251 non-null  int64
 6   Marital_Status    11251 non-null  int64
 7   State             11251 non-null  object
 8   Zone              11251 non-null  object
 9   Occupation        11251 non-null  object
 10  Product_Category  11251 non-null  object
 11  Orders            11251 non-null  int64
 12  Amount            11239 non-null  float64
 13  Status            0 non-null      float64
 14  unnamed1          0 non-null      float64
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB
```

df.drop(['Status','unnamed1'], axis=1, inplace=True)df.info()pd.isnull(df) 2.data preprocessing

checking for the null values

[10]:
```
pd.isnull(df).sum()
```

[10]:
```
User_ID              0
Cust_name            0
Product_ID           0
Gender               0
Age Group            0
Age                  0
Marital_Status       0
State                0
Zone                 0
Occupation           0
Product_Category     0
Orders               0
Amount              12
dtype: int64
```

[11]:
```
df.shape
```

[11]:
```
(11251, 13)
```

[12]:
```
df.dropna(inplace=True)
```

[13]:
```
df.shape
```

[13]:
```
(11239, 13)
```

[14]:
```
df['Amount'] = df['Amount'].astype('int')
```

[15]:
```
df['Amount'].dtypes
```

[15]:
```
dtype('int64')
```

[16]:
```
df.columns
```

[16]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

[17]:
```
df.rename(columns={'Marital_Status' : 'Shadi'})
```

[17]:
```
        User_ID    Cust_name Product_ID Gender Age Group  Age  Shadi  \
0       1002903    Sanskriti  P00125942      F     26-35   28      0
1       1000732       Kartik  P00110942      F     26-35   35      1
2       1001990        Bindu  P00118542      F     26-35   35      1
3       1001425       Sudevi  P00237842      M      0-17   16      0
4       1000588         Joni  P00057942      M     26-35   28      1
…         …          …        …    …       …  …    …
11246   1000695      Manning  P00296942      M     18-25   19      1
11247   1004089  Reichenbach  P00171342      M     26-35   33      0
11248   1001209        Oshin  P00201342      F     36-45   40      0
11249   1004023       Noonan  P00059442      M     36-45   37      0
11250   1002744      Brumley  P00281742      F     18-25   19      0

                 State      Zone       Occupation Product_Category  Orders  \
0          Maharashtra   Western       Healthcare             Auto       1
1       Andhra Pradesh  Southern             Govt             Auto       3
2        Uttar Pradesh   Central       Automobile             Auto       3
3            Karnataka  Southern     Construction             Auto       2
4              Gujarat   Western  Food Processing             Auto       2
…                 …       …              …              …      …
11246      Maharashtra   Western         Chemical           Office       4
11247          Haryana  Northern       Healthcare       Veterinary       3
11248   Madhya Pradesh   Central          Textile           Office       4
11249        Karnataka  Southern      Agriculture           Office       3
11250      Maharashtra   Western       Healthcare           Office       3

        Amount
0        23952
1        23934
2        23924
3        23912
4        23877
…        …
11246      370
```

```
11247     367
11248     213
11249     206
11250     188

[11239 rows x 13 columns]
```

[18]:
```
df.describe()
```

[18]:

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

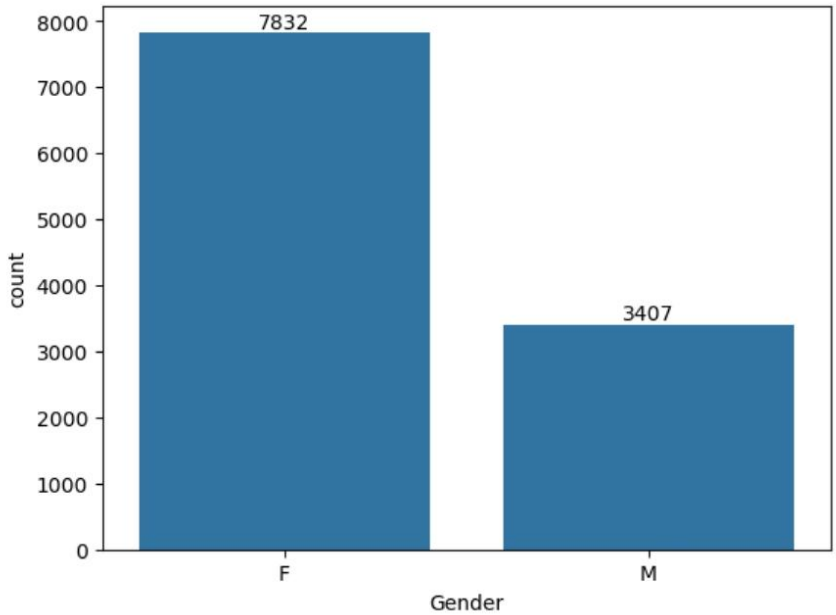
[19]:
```
df[['Age','Orders','Amount']].describe()
```

[19]:

	Age	Orders	Amount
count	11239.000000	11239.000000	11239.000000
mean	35.410357	2.489634	9453.610553
std	12.753866	1.114967	5222.355168
min	12.000000	1.000000	188.000000
25%	27.000000	2.000000	5443.000000
50%	33.000000	2.000000	8109.000000
75%	43.000000	3.000000	12675.000000
max	92.000000	4.000000	23952.000000

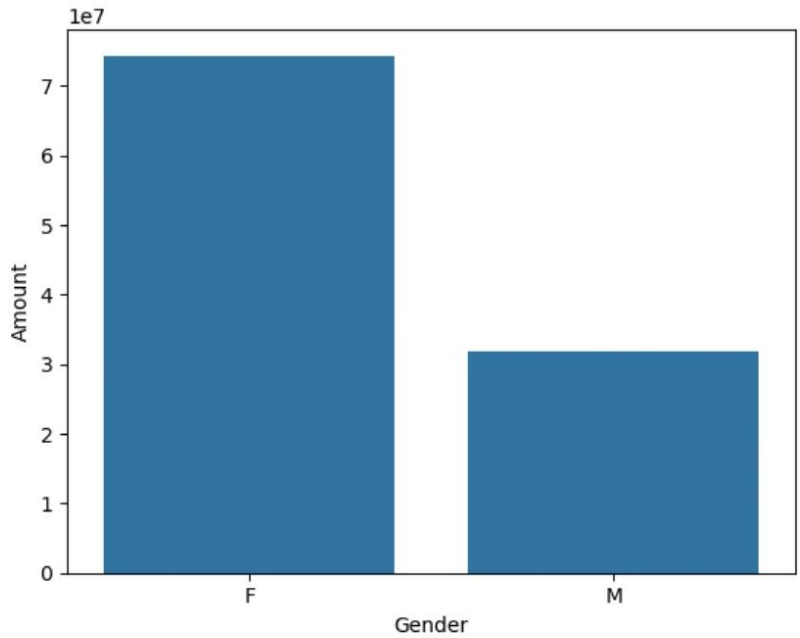
3. Data Analysis and Visualisation

[20]:
```
ax = sns.countplot(x='Gender', data = df)

for bars in ax.containers:
    ax.bar_label(bars)
```

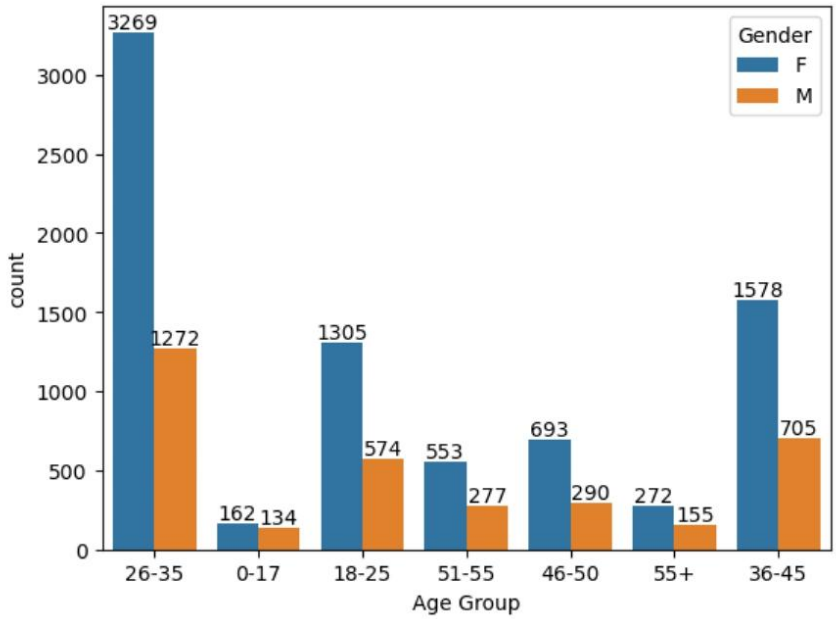


[21]:
```
sales_gen = df.groupby(['Gender'], as_index = False)['Amount'].sum().
 ↪sort_values(by='Amount', ascending = False)
sns.barplot(x= 'Gender',y= 'Amount', data =sales_gen)
```

[21]: <Axes: xlabel='Gender', ylabel='Amount'>

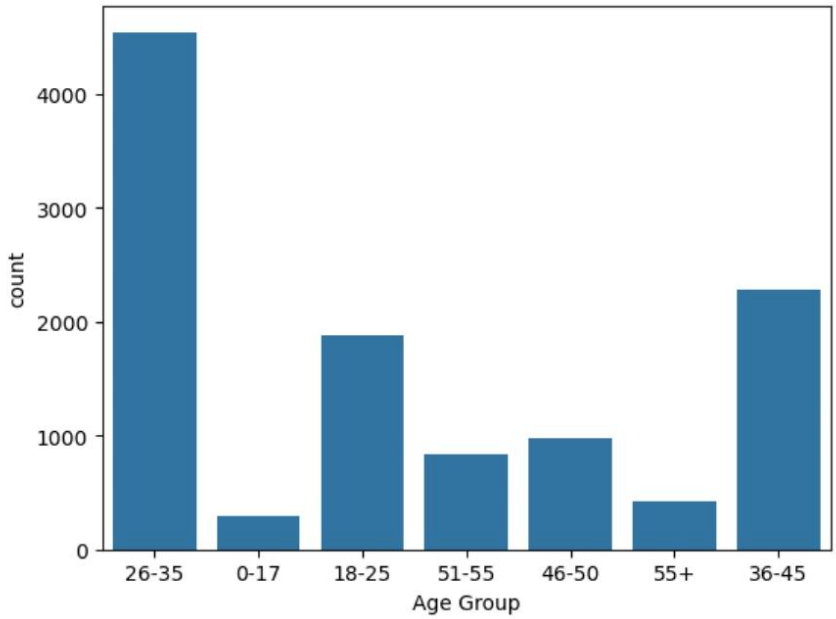


[23]:
```
ax=sns.countplot(data=df , x= 'Age Group', hue = 'Gender')

for bars in ax.containers:
    ax.bar_label(bars)
```



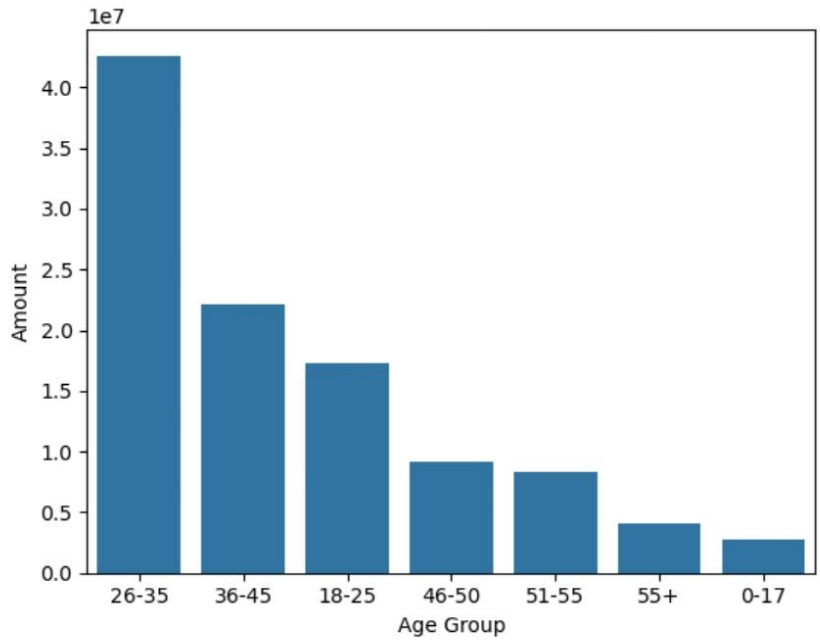
[25]:
```
sns.countplot(data=df, x= 'Age Group')
```

[25]: <Axes: xlabel='Age Group', ylabel='count'>



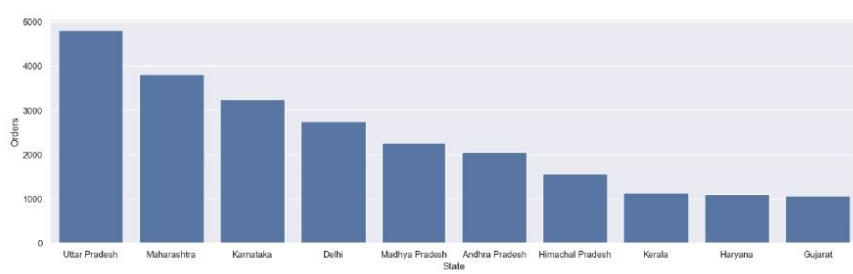
[26]:
```
sales_age= df.groupby(['Age Group'], as_index = False)['Amount'].sum().
 ↪sort_values(by='Amount',ascending =False)

sns.barplot(x='Age Group', y = 'Amount', data= sales_age)
```

[26]: <Axes: xlabel='Age Group', ylabel='Amount'>



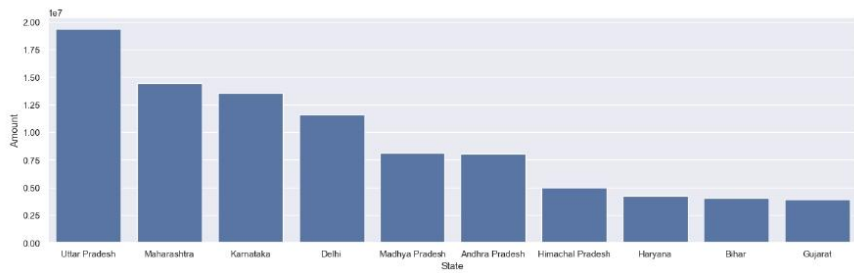
[27]:
```
sales_state = df.groupby(['State'], as_index=False)['Orders'].sum().
↪sort_values(by='Orders', ascending=False).head(10)

sns.set(rc={'figure.figsize':(18,5)})

sns.barplot(data= sales_state, x='State', y= 'Orders')
```

[27]: <Axes: xlabel='State', ylabel='Orders'>

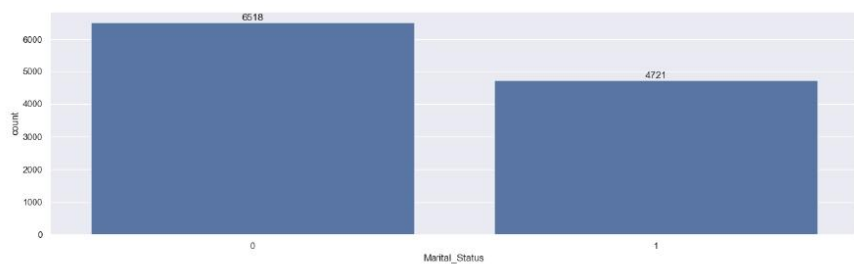


[30]:
```
sales_state = df.groupby(['State'], as_index=False)['Amount'].sum().
↪sort_values(by='Amount', ascending=False).head(10)

sns.set(rc={'figure.figsize':(18,5)})
sns.barplot(data = sales_state, x = 'State',y= 'Amount')
```

[30]: <Axes: xlabel='State', ylabel='Amount'>

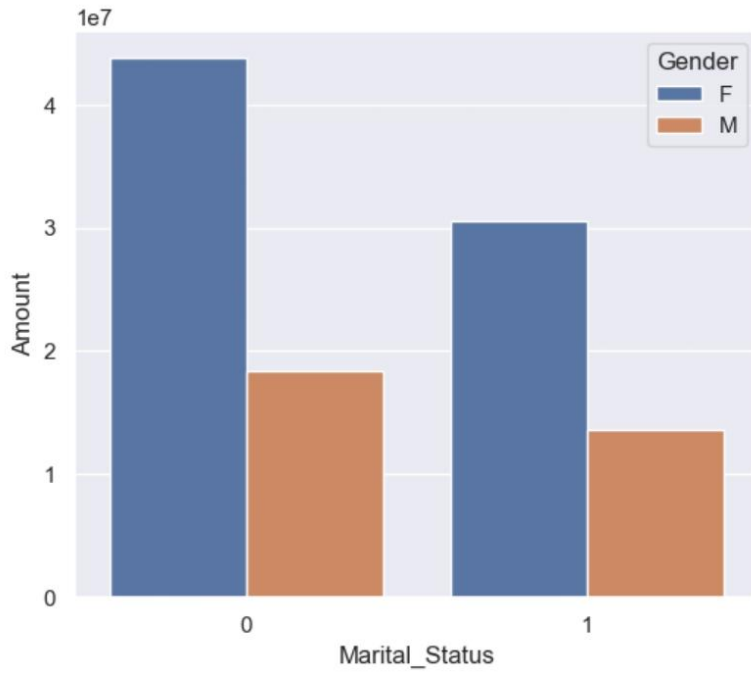


[31]:
```
ax = sns.countplot(data = df, x = 'Marital_Status')

sns.set(rc={'figure.figsize':(17,5)})
for bars in ax.containers:
    ax.bar_label(bars)
```



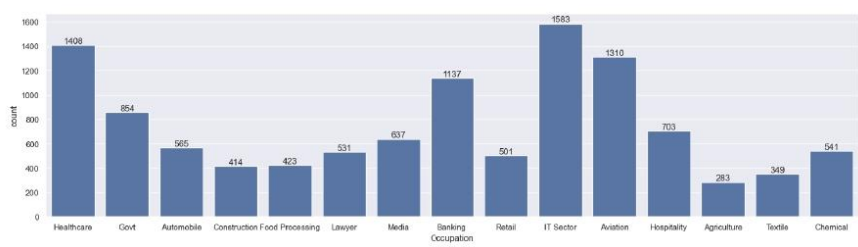
[32]:
```
sales_state = df.groupby(['Marital_Status', 'Gender'],␣
↪as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)

sns.set(rc={'figure.figsize':(6,5)})
sns.barplot(data = sales_state, x = 'Marital_Status',y= 'Amount', hue='Gender')
```

[32]: <Axes: xlabel='Marital_Status', ylabel='Amount'>



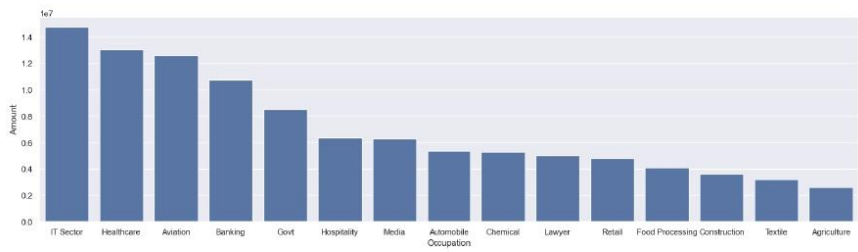
[42]:
```
sns.set(style="darkgrid")
sns.set_palette("deep")
```

[43]:
```
sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Occupation')

for bars in ax.containers:
    ax.bar_label(bars)
```

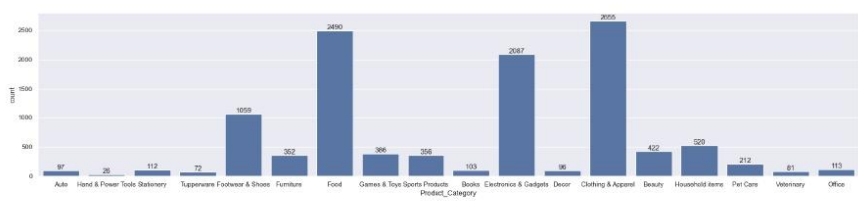


[44]:
```
sales_state = df.groupby(['Occupation'], as_index=False)['Amount'].sum().
 ↪sort_values(by='Amount', ascending=False)

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Occupation',y= 'Amount')
```

[44]: <Axes: xlabel='Occupation', ylabel='Amount'>

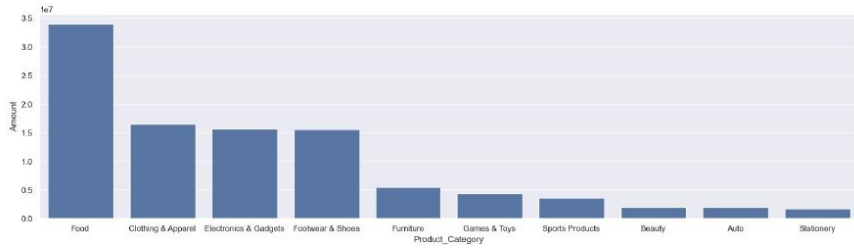


[45]:
```
sns.set(rc={'figure.figsize':(25,5)})
ax = sns.countplot(data = df, x = 'Product_Category')

for bars in ax.containers:
    ax.bar_label(bars)
```



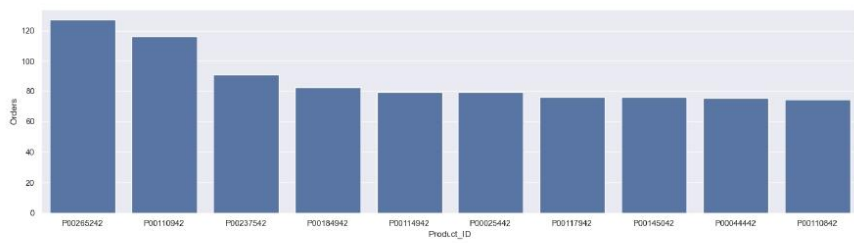
[46]:
```
sales_state = df.groupby(['Product_Category'], as_index=False)['Amount'].sum().
 ↪sort_values(by='Amount', ascending=False).head(10)

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_Category',y= 'Amount')
```

[46]: <Axes: xlabel='Product_Category', ylabel='Amount'>



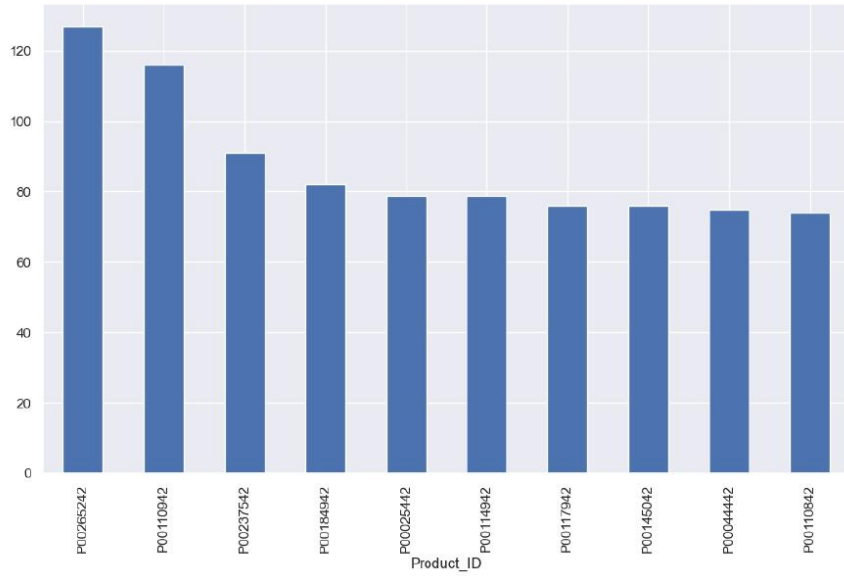
[47]:
```
sales_state = df.groupby(['Product_ID'], as_index=False)['Orders'].sum().
 ↪sort_values(by='Orders', ascending=False).head(10)

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_ID',y= 'Orders')
```

[47]: <Axes: xlabel='Product_ID', ylabel='Orders'>



[48]:
```
fig1, ax1 = plt.subplots(figsize=(12,7))
df.groupby('Product_ID')['Orders'].sum().nlargest(10).
 ↪sort_values(ascending=False).plot(kind='bar')
```

[48]: <Axes: xlabel='Product_ID'>



[50]:
```
df.shape
```

[50]: (11239, 13)

Conclusion

The Diwali Sales Analysis provided valuable insights into customer purchasing behavior during the festive season.

The analysis showed that female customers and the 26–35 age group contribute the highest share of total sales.

Certain states, occupations, and product categories play a major role in driving festive revenue.

Data visualization and exploratory analysis helped transform raw sales data into meaningful business insights.

Overall, the study demonstrates how data analytics can support better decision-making and improve festive sales performance.

Future Scope

The analysis can be extended by including Diwali sales data from multiple years for trend comparison.

Predictive models can be applied to forecast future festive sales.

Online and offline sales data can be analyzed separately for deeper insights.

Customer feedback and marketing campaign data can be integrated for advanced analysis.

[ ]:

```

```

RECOMMENDATIONS

Based on analytical findings, businesses should adopt demographic-focused marketing strategies.

Inventory planning should align with regional demand concentration. Festive discount campaigns can target high-spending age groups.

Loyalty programs can retain repeat customers beyond festive season.

Personalized digital advertising should target economically stable occupation groups.

SYSTEM ENVIRONMENT

The Diwali Sales Analysis project was developed and executed in a standard computing environment suitable for data analysis tasks. The hardware requirements for the system include a minimum of 4GB RAM, an Intel i3 (or equivalent) processor, and sufficient storage to handle dataset files and program execution. Since the dataset size is moderate, no advanced hardware infrastructure or high-performance computing system was required.

The software environment consists of Windows operating system along with Python 3.x as the primary programming language. Jupyter Notebook was used as the integrated development environment (IDE) to write, execute, and document the analysis code in an interactive manner. The analysis relied on several Python libraries including Pandas for data manipulation and cleaning, NumPy for numerical computations, Matplotlib and Seaborn for data visualization and graphical representation.

The system environment ensures smooth execution of data preprocessing, exploratory data analysis, and visualization tasks. The setup is cost-effective, user-friendly, and easily reproducible, making it suitable for both academic and small-scale business analytics projects.

LIMITATIONS

The dataset represents only one festive season, limiting longitudinal trend evaluation.

Income levels were not available, restricting economic-depth segmentation.

The analysis is exploratory and does not incorporate machine learning models.

External factors such as marketing campaigns or economic conditions were not considered.

FUTURE SCOPE

Future scope includes implementation of predictive sales forecasting models using regression techniques.

Clustering algorithms can segment customers more precisely.
Multi-year comparison can identify evolving trends.

Integration with dashboard systems such as Power BI can provide real-time insights.

Advanced analytics can convert this project into a full retail intelligence system.

CONCLUSION

The Diwali Sales Analysis project successfully demonstrates how data analytics techniques can be applied to understand festive season retail performance.
By analyzing over 11,000 transaction records, the study identified important patterns related to customer demographics, regional sales distribution, occupation-based purchasing behavior, and product category performance.

Through systematic data preprocessing, the dataset was cleaned and structured to ensure accuracy in analysis. Exploratory Data Analysis (EDA) was performed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn to extract meaningful insights. The analysis revealed that female customers contributed a larger share of total revenue during the Diwali season.
Additionally, customers in the 26–35 age group emerged as the most active and highest spending demographic segment. Regional evaluation highlighted that certain states generated significantly higher festive sales, indicating geographic concentration of purchasing power.

The findings emphasize the importance of data-driven decision-making in retail management. Instead of relying on assumptions, businesses can use analytical methods to identify target customers, optimize inventory,

and design more effective marketing campaigns during festive seasons. Overall, the project provides practical exposure to real-world retail data analysis and highlights the role of analytics in improving business strategies. It serves both as an academic study and as a demonstration of how structured data analysis can support better decision-making in the retail industry.

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